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Paul Colombo

COMPUTER SCIENCE & ROBOTICS · CARNEGIE MELLON '28

EDUCATION

Carnegie Mellon University · School of Computer Science — Pittsburgh, PA
B.S. Computer Science & Robotics — Expected 2028

EXPERIENCE

Software Engineering Intern · Common Good Ed — Summer 2026

- Built a document-review workflow that lets reviewers vet any material before it enters the curriculum.
- Designed a Firebase datastore that logs who reviewed which document, creating an auditable paper trail that keeps a human in the loop so AI-generated errors are caught before they reach students.

Undergraduate Researcher · Temel Lab, CMU Robotics Institute — 2025 – Present

- Built computer-vision software that tracks animal trajectories from raw video, working alongside PhD students in Prof. Zeynep Temel's lab.

Data Acquisition Lead Engineer · Carnegie Mellon Racing — Fall 2025 – Present

- Lead for DAQx, our own data analysis software. Along with live telemetry software work.
- Build and calibrate sensor systems that capture aero, tire-heat, and speed telemetry during testing and races.
- Read electronics schematics and programmed the sensor Raspberry Pi on the car.

Student Engineer · MIT Lincoln Laboratory (LLRISE) — July 2023

- Selected as one of 22 students for a residential engineering program at MIT Lincoln Laboratory.
- Built, coded, and tested Doppler and range radar hardware from the ground up.

Student Developer · AP CSP Course Website — 2022

- One of three students chosen by the CS department to build a course website from scratch.
- Used Node.js, HTML, CSS, and JavaScript with a backing database to store student grades; the site is now used by Cambridge Rindge & Latin to teach AP CS Principles.

PROJECTS

Autonomous Mapping Drone · Personal build — Jetson Orin Nano · Unitree L2

- Built a drone from scratch that maps terrain and localizes itself, using lidar and without GPS, in real time.

BeeTracker · Temel Lab research tool — YOLO · ByteTrack

- Built a video pipeline that tracks bee trajectories and counts hive entries/exits, replacing thousands of hand-drawn bounding boxes with an afternoon of automated labeling.
- Combined a fine-tuned YOLO (neural network) model with ByteTrack and an iterative autolabel loop (seed → audit → train → relabel) holding ~0.86 mean IoU against reviewer-corrected ground truth; exports tracks to CSV/JSON.

TECHNICAL SKILLS

Languages	Java, JavaScript, C/C++, Python, SML
Web	React, Node.js, Vite, Bootstrap, HTML/CSS, SQL, Firebase
ML / Robotics	PyTorch, sensor fusion, LiDAR, gyro/accelerometer, Arduino, CAD, soldering
Tools	Git, Linux, AWS, Gradle, LaTeX, JavaFX, Kali, VirtualBox

RELEVANT COURSEWORK

General Robotics · Robotics Kinematics and Dynamics · Sensing and Sensors · Feedback & Control Systems · Intro to Computer Systems · Parallel & Sequential Algorithms · Probability Theory · Concepts in AI · Principles of Functional Programming · Great Ideas in Theoretical CS · Imperative Computation · Matrices & Linear Transformations · Vector Calculus

ADDITIONAL

CS Tutor (2025) · **Automotive Technician**, Liberty Automotive & Repair (2023–25) · **Farmhand**, Sylvester Manor Educational Farm (2022)